Evolution and Validation of Practice Standards, Training, and Professional Development

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Learning Objectives

1. Describe landmark events in the evolution of critical care pharmacy as a specialty, including summarizing key published documents and primary evidence validating critical care pharmacy as a specialty.
2. Identify the core knowledge areas for pharmacists caring for critically ill patients.
3. Outline criteria for credentialing, conventional training pathways and mentorship, and standards of practice for clinical pharmacy in the critical care practice environment..
4. Develop an approach to conducting a gap analysis relative to the principles and values of team-based care in a local critical care practice environment.
5. Develop an approach to lifelong professional learning to maintain competency in critical care pharmacy practice using the principles of continuing personal professional development.
6. Identify the avenues and processes for contributing to the critical care body of knowledge as a presenter, author, or peer reviewer.

Abbreviations in This Chapter

ACCM	American College of Critical Care Medicine
ACCP	American College of Clinical Pharmacy
ACLS	Advanced cardiac life support
ASHP	American Society of Health-System Pharmacists
BPS	Board of Pharmacy Specialties
CE	Continuing education
CPD	Continuing professional development
CPE	Continuing pharmacy education
ICU	Intensive care unit
PDP	Personal development plan
SCCM	Society of Critical Care Medicine

Self-Assessment Questions

Answers and explanations to these questions can be found at the end of the chapter.

1. Which is the journal that, in 1982, was the first to publish a critical care therapeutics column?
   A. *Drug Intelligence and Clinical Pharmacy.*
   B. *Pharmacotherapy.*
   C. *Chest.*
   D. *Heart & Lung.*

2. The scope of pharmacy practice within the intensive care unit (ICU) was outlined by two task forces focused on models of critical care delivery, the definition of an intensivist, and the practice of critical care medicine within three different proposed models in 2001. Which best depicts the professional organization that formed these two task forces?
   A. National Academy of Medicine (formerly Institute of Medicine).
   B. American College of Clinical Pharmacy (ACCP).
   C. American College of Critical Care Medicine (ACCM).
   D. Clinical Pharmacy and Pharmacology Section of the Society of Critical Care Medicine (SCCM).

3. Which best depicts the medical event that has been documented to be associated with a lower mortality in ICUs with clinical pharmacists compared with ICUs without clinical pharmacists?
   A. Corrected QT (QTc)-interval prolongation.
   B. Preventable adverse drug interactions.
   C. Drug-drug interactions.
   D. Thromboembolic/infarction-related events.

4. Which best describes what is deemed a core knowledge base area for pharmacists caring for critically ill patients?
   A. Nephrology.
   B. Dermatology.
   C. Rheumatology.
   D. Obstetrics.

5. Which most accurately reflects the journal that published a landmark study documenting a decrease in preventable adverse drug reactions after the inclusion of pharmacists on interdisciplinary medical rounds?
   A. *New England Journal of Medicine.*
   B. *Lancet.*
   C. *Journal of the American Medical Association.*
   D. *Annals of Internal Medicine.*

6. There were eight American Society of Health-System Pharmacists (ASHP)-accredited critical care pharmacy residencies in 2001. Which annual absolute value represents the approximate increase in ASHP-accredited critical care residencies between 2001 and 2021?
   A. 5.
   B. 8.
   C. 12.
   D. 15.

7. Which best reflects the current conventional or preferred postgraduate training pathway to clinical pharmacy practice in an ICU providing level I services?
   A. Postgraduate year 1 (PGY1) residency with focused critical care rotations.
   B. PGY1 residency followed by on-the-job mentored training.
   C. PGY1 and PGY2 critical care residency.
   D. PGY1 and critical care traineeship.

8. When considering the five principles of team-based health care delineated in the Institute of Medicine (now National Academy of Medicine) discussion paper, which of the other four principles is effective communication most tightly linked to?
   A. Shared goals.
   B. Clear roles.
   C. Mutual trust.
   D. Measurable processes and outcomes.

9. Which statement is most accurate concerning the mentor-mentee relationship as it pertains to the training and development of critical care pharmacists?
   A. Formal mentoring relationships are restricted to residency and fellowships.
   B. Voluntary relationships that evolve and develop through mutual interests have the greatest likelihood of success.
   C. Mentored training programs are the only reliable pathway for the nonconventional training of critical care pharmacists.
   D. Most successful critical care pharmacists have a single relevant mentor-mentee relationship during their training and development.

10. Which is the most accurate description of the relationship between the continuing pharmacy education (CPE) and the continuing professional development (CPD) of clinical pharmacists?
    A. CPE and CPD are two distinctly different processes for continuing development.
    B. CPD is an individualized, self-directed, and iterative process of development that replaces traditional CPE.
    C. CPE is strictly a didactic process, whereas CPD incorporates many different learning strategies and techniques.
    D. CPD is an individualized, self-directed process that typically incorporates relevant CPE as one of the learning strategies.

11. Which statement is most accurate relative to the recently published standards of care and standardized process of care for clinical pharmacy when considering critical care pharmacy practice?
    A. ICUs are highly individualized practice environments that cannot easily conform to broad-based, discipline-wide standards.
    B. Critical care pharmacists have knowledge and skill sets that are specific to their practice style and environment and are not consistent with the standards.
    C. The standards of care and standardized process of care are very consistent with critical care pharmacy practice standards and expectations.
    D. The standard process of care, which has evolved around the "provider status" efforts, is primarily applicable to the ambulatory, primary care environment of practice.

12. Which is the best example of an audience that has not traditionally been an important focus of critical care pharmacists' educational and teaching efforts?
    A. Patients and families.
    B. Pharmacy students.
    C. Critical care physicians.
    D. Critical care nurses.

13. As part of the reflection stage of the CPD process, which would be the best example of an episodic opportunity for self-assessment?
    A. An annual 360-degree peer evaluation providing feedback that your coworker believes you do not contribute adequately on departmental initiatives.
    B. Recognition that the usual approach to training and assessing a challenging student on rotation was ineffective.
    C. A self-evaluation of the past year's accomplishments for your direct supervisor.
    D. An annual performance evaluation with specific goals for the coming year.

14. Which would be considered the most valid reason for recommending the rejection of a manuscript as a scientific reviewer?
    A. Poor syntax and word choices.
    B. A methodological flaw that results in incorrect data.
    C. A disagreement concerning the statistical analysis presented in the manuscript.
    D. Results presented in the abstract that are inconsistent with results presented in a table.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 2: Therapeutic and Patient Management
   a. Task C: 5
2. Domain 3: Professional Practice
   a. Task A: 3, 7
   b. Task B: 1, 2, 4, 5

I. LANDMARK EVENTS IN CRITICAL CARE MEDICINE/PHARMACY

A. First ICU: Three-Bed Neurosurgical Unit in Baltimore, Maryland (1930s) (Ann Pharmacother 2006;40:612-8)

B. First Pharmacists Assigned to ICUs in a Limited Number of Hospitals – Occurred in the late 1960s (Ann Pharmacother 2006;40:612-8)

C. Establishment of several Critical Care Pharmacists ICU Practices (in or around the 1970s; clinical research conducted in a wide array of therapeutic specialty areas [e.g., pharmacokinetics, infectious diseases, nutrition support, ACLS]) (Ann Pharmacother 2006;40:612-8; Practice of Critical Care Pharmacy, Rockville, MD: Aspen Publications, 1985)
   1. Cardiovascular ICUs
   2. Pediatric/neonatal ICUs
   3. Medical ICUs
   4. Emergency medicine
   5. Trauma
   6. Surgical ICUs
   7. Neurosurgical ICUs

D. Formation of SCCM with 100 Members (1970)
   1. Multidisciplinary model stressed by founding SCCM President Max Harry Weil, M.D.
   2. Subsequent inclusion of pharmacists as permanent members of the SCCM governing council

E. Emergence of Critical Care Specialty Journals and Publications
   1. *Heart & Lung* (1972)
   2. *Intensive Care Medicine* (1972)
   3. *Critical Care Medicine* (1973)
   4. *Critical Care Clinics* (1984)
   5. Critical care therapeutics column in *Drug Intelligence and Clinical Pharmacy* (1982)
   6. First critical care pharmacy textbook: *Critical Care Pharmacy* (1985)

F. Formation of Critical Care Pharmacists Specialty Groups
   1. SCCM Clinical Pharmacy and Pharmacology Section (1989)
   2. ACCP Critical Care Practice and Research Network (PRN) (1992)
   3. Neurocritical Care Society (NCS), Pharmacy Section (circa 1999)

II. VALIDATION OF CRITICAL CARE PHARMACY AS A SPECIALTY

A. Chapter Titled "Role of the Pharmacist in Caring for the Critically Ill Patient," Published in The Pharmacologic Approach to the Critically Ill Patient, 3rd ed. Baltimore, MD: Williams & Wilkins, 1994:156-66.

B. Publication of "Position Paper on Critical Care Pharmacy Services" (Crit Care Med 2000;28:3746-50); establishment of three levels of pharmacy services (fundamental, desirable, optimal) for the provision of pharmaceutical care in critically ill patients; major update in 2020 with recommendations simplified to essential and desirable. Statements and recommendations categorized as follows: Patient Care (n=34), Quality Improvement (n=21), Research/Scholarship (n=9), Training Education (n=10), and Professional Development (n=8) (Crit Care Med 2020;48:1375-82) (see Table 1 for details).

C. SCCM Recognition of Critical Care Pharmacists as an Integral Member of the Multidisciplinary Team Together with Physicians, Nurses, and Respiratory Therapists (Crit Care Med 2001;29:2007-19)

D. Scope of Pharmacy Practice Within ICU Outlined by Two ACCM Task Forces: Models of critical care delivery and definition of an intensivist and the practice of critical care medicine within three different proposed models (critical care pharmacy and pharmacist services deemed "essential" within level I critical care centers as endorsed by ACCM (Crit Care Med 2003;31:2677-83) (see specifics that follow)

E. Awarding of FCCM Status by ACCM: 111 pharmacists from an overall total of around 1399; nine of total 89 MCCMs (Master Critical Care Medicine) (2021); awarding FNCS status by the Neurocritical Care Society (n=8) Receipt of Several Honors and Awards by Critical Care Pharmacists Within SCCM, ACCM, ACCP, and NCS
   1. SCCM/ACCM: Distinguished Investigator Award, Shubin-Weil Master Clinician, Excellence in Bedside Teaching Award, Distinguished Service Award
   2. ACCP: Russell R. Miller Award, Paul F. Parker Medal, Clinical Practice Award, Robert M. Elenbaas Service Award, Education Award
   3. Creation of the Joseph F. Dasta Critical Care Pharmacy Outcomes Research Grant by the Clinical Pharmacy and Pharmacology Section of SCCM (2000); Barry A. Shapiro Memorial Award for Excellence in Critical Care Management

F. Leadership Roles in Major Multidisciplinary and Pharmacy Organizations
   1. SCCM: Presidents Judith Jacobi (2010), Sandra L. Kane-Gill (2022)
   2. ACCP: Presidents Robert Elenbaas, Barbara Zarowitz, Bradley Boucher, Curtis Haas, Judith Jacobi, Brian Erstad, Jimmi Hatton Kolpek
   3. Neurocritical Care Society: President Gretchen Brophy (2017-2018)
   4. ASHP: President: Thomas J. Johnson (2020)

G. Standards for Neurologic Critical Care Units: Inclusion of dedicated pharmacists on the neurocritical care team (Neurocrit Care 2018;29:145-60)

III. CRITICAL CARE PHARMACY GROWTH

A. SCCM Membership: Over 16,000 total; over 2300 with pharmacy credentials. ACCP Critical Care PRN: over 1600 (June 2023)

B. Board Certified Critical Care Pharmacists (BCCCPs): Around 3994 (2022 BPS Annual Report [http://books.bpsweb.org/books/jpjn/#p=26])

C. Pharmacist Provision of Direct Patient Care Services: 62.2% of ICUs responding to hospital survey sent to pharmacy directors identified by the American Hospital Association as having an ICU pharmacist (Ann Pharmacother 2006;40:612-8)

IV. STUDIES DOCUMENTING THE ASSOCIATION OF CRITICAL CARE PHARMACY SERVICES WITH FAVORABLE HEALTH CARE OUTCOMES

A. Reduction in Drug Costs in ICU with the Inclusion of a Pharmacist as a Member of the Multidisciplinary Team
   1. Medical-surgical ICU: Annual savings of $67,664 in 1994 dollars (Crit Care Med 1994;22:1044-8)
   2. Several other studies in a wide range of ICU settings (see Intensive Care Med 2003;29:691-8)
   3. Burn ICU: Annual savings of $22,162 in 2003 dollars (J Burn Care Res 2006;310-13)
   4. Neurosurgical ICU: Reduction in pharmacy acquisition costs from $4833 to $3239 per patient after the addition of a pharmacist to the neurosurgery team; reduction in ICU days from 8.56 to 7.24 days (p=0.003) (Neurosurgery 2009;65:946-50; discussion 950-1)
   5. Scoping review conducted for cost avoidance generated by clinical pharmacists on interventions performed in and ICU or emergency department identified 38 distinctive categories (Pharmacotherapy 2019;39:215-31). Across 55,926 interventions, estimated cost avoidance was $418.48 per intervention and $845.49 per patient. Overall ratio of cost avoidance/pharmacist salary was between $3.3:1 and $9.6:1 (Crit Care Explor 2021;3:e0594).
   6. Multicenter, observational cohort study in critically ill adult patients across a 25-hospital integrated health care system evaluating telehealth critical care pharmacist services in 8-hour shifts, 7 days a week. Found pharmacists documented 2838 interventions associated with $1,664,254 gross cost avoidance over 2 years. Expense:cost avoidance ratio was 4.5:1 (Crit Care Explor 2023;5:e0839).

B. Reduction in Adverse Drug Effects/Drug-Drug Interactions
   1. Decrease in preventable adverse drug effects after the inclusion of a pharmacist on interdisciplinary medical ICU rounds: 66%, ($p<0.001$) (JAMA 1999;282:267-70); supported by meta-analysis of three studies (odds ratio (OR) 0.23, 95% confidence interval (0.11, 0.48) (J Crit Care 2015;30:1101-06). Reduced; preventable and nonpreventable adverse drug events (OR 0.26, 95% CI (0.15, 0.44), $p<0.0001$ and OR 0.47, 95% CI (0.28, 0.77, p=0.003, respectively) (Crit Care Med 2019;47:1243-50).
   2. Decreased incidence of QTc-interval prolongation with ICU monitoring by a pharmacist using a standard algorithm: 19% versus 39% (p=0.006) (Ann Pharmacother 2008;42:475-82)
   3. Reduction in drug-drug interactions by 65% ($p<0.01$) in medical ICUs with a pharmacist (J Crit Care 2011;26:104.e101-106)

C. Improvement in Infectious Diseases Morbidity, Mortality, and Costs (Crit Care Med 2008;36:3184-9)
   1. Mortality in ICUs without clinical pharmacists than in ICUs with clinical pharmacists: 23.6% higher for nosocomial-acquired infections in ICUs without clinical pharmacists ($p<0.001$), 16.2% higher for community-acquired infections in ICUs without clinical pharmacists (p=0.008), 4.8% higher for sepsis in ICUs without clinical pharmacists ($p\leq0.008$)
   2. Longer length of stay for ICUs without clinical pharmacists than for ICUs with clinical pharmacists: 7.9% for nosocomial-acquired infections in ICUs without clinical pharmacists ($p<0.001$), 5.9% for community-acquired infections in ICUs without clinical pharmacists (p=0.03), 8.1% for sepsis in ICUs without clinical pharmacists ($p<0.001$)
   3. Increased Medicare billing in ICUs without clinical pharmacists compared with ICUs with clinical pharmacists: 12% for nosocomial-acquired infections in ICUs without clinical pharmacists, 11.9% for community-acquired infections in ICUs without clinical pharmacists, 12.9% for sepsis in ICUs without clinical pharmacists ($p<0.001$)

D. Improvement in Thromboembolic and Infarction-Related Event (TIE) Clinical and Economic Outcomes (Pharmacotherapy 2009;29:761-8)

1. Mortality increased in ICU patients with TIE without clinical pharmacy services compared with ICU patients with clinical pharmacy services: 37%, ($p<0.0001$).
2. Bleeding complications increased by 49% ($p<0.001$), with 39% more patients receiving transfusions ($p=0.006$) in ICUs without clinical pharmacy services.
3. Length of ICU stays and costs were significantly higher in patients with TIE in ICUs without clinical pharmacy services.

E. Meta-Analysis of 14 Studies of Critical Care Pharmacists as Member of Multidisciplinary Team (Crit Care Med 2019;1243-50)
   1. Reduced mortality: OR 0.78, 95% CI (0.73-0.83, $p<0.00001$)
   2. LOS reduction: mean -1.33 days, 95% CI (-1.75, -0.90), $p<0.00001$)

F. Impact of ICU Protocols on Patient Outcomes
   1. Significant improvement in sedation and analgesia monitoring targets with the use of protocol versus empiric therapy ($p\leq0.01$); no difference in length of ICU stay (Pharmacotherapy 2000;20:662-72)
   2. Pharmacist-enforced ICU sedation protocol reduced mechanical ventilator duration as well as ICU and hospital length of stay ($p<0.001$) (Crit Care Med 2008;36:427-33)
   3. Improvement Mortality with Inclusion of Clinical Pharmacist Following Multicomponent Intervention in Tertiary Care Medical ICU (also included increase in ICU beds, larger rooms, 24-hour critical care specialist coverage, decrease in respiratory therapist/patient ratio) (Crit Care Med 2011;39:284-93)
      a. ICU Mortality decrease from 18.4% to 14.9% ($p=0.006$), hospital mortality decrease 25.8% to 21.7% ($p=0.005$)
      b. Increase in median ICU length of stay; no difference in hospital length of stay
      c. Increase in median 28-day ventilator-free days in mechanically ventilated patients
      d. Mean decrease in daily dosing of fentanyl and lorazepam
   4. Pharmacist management of pain, agitation, and delirium in ICU through multidisciplinary bundle; 46% reduction in continuous sedation, reduction in ICU and hospital length of stay (Am J Health-Syst Pharm 2017;74:253-62); systematic review of pharmacist-led interventions on pain, agitation, and delirium in mechanically ventilated adults (J Am Coll Clin Pharm 2023;6:1041-52).
   5. Pharmacist involvement in multidisciplinary initiative to reduce sepsis-related mortality in ICU (i.e., "Code Sepsis"); reduction in mean time from sepsis screen to antibiotic administration from 427 minutes to 31 minutes (Am J Health-Syste Pharm 2016;73(3):143-9)
   6. Leadership role in developing clinical practice guidelines for preventing and managing pain, agitation/sedation, delirium, immobility, and sleep disruption in adult critically ill patients (PADIS Guidelines) (Crit Care Med 2002;30:119-41; Crit Care Med 2018;46:e825-73) and clinical practice guidelines for sustained neuromuscular blockade in critically ill adults (Crit Care Med 2016;44:2079-103).
   7. Before and after study of critical care pharmacist interventions on drug therapy and clinical strategies: Reduction in hospital length of stay (mean 3.7 days, $p<0.001$), ICU length of stay (1.4 days, $p<0.01$), duration of mechanical ventilation (1.2 days, $p<0.01$), hospital costs per stay (2560 euros, $p<0.001$); no impact on mortality. (Crit Care Med 2018;46:199-207)
   8. Retrospective cohort study of albumin use in critically ill patients at single academic medical center: Reduction in inappropriate albumin use (50.9%, $p<0.001$), total annual cost savings greater than $355,000. (Ann Pharmacotherapy 2020;54:105-112).

V. PRACTICE STANDARDS FOR CRITICAL CARE PHARMACY

A. Standards of Practice for Clinical Pharmacists (J Am Coll Clin Pharm 2023;6:1156-9): The Standards of Practice for Clinical Pharmacists were published by ACCP, incorporating a standardized process of care endorsed by all major pharmacy organizations. This document defines expectations of clinical pharmacists delivering comprehensive medication management in team-based, collaborative practice settings, including the ICU.
   1. Qualifications
      a. Licensed pharmacists
      b. Advanced education, training, and experience
         i. Advanced, accredited residency in critical care pharmacy (PGY2);
         ii. Fellowship in critical care research; or
         iii. Equivalent, relevant clinical experience
      c. Clinical and personal competencies to practice in a team-based collaborative environment
      d. Board certification
   2. Process of care
      a. Collection of patient information
      b. Assessment of medication regimen
      c. Development of a plan of care for medication optimization
      d. Implementation of the plan of care
      e. Follow-up evaluation and medication monitoring
   3. Documentation
      a. Medication history
      b. Active problem list with assessment
      c. Plan of care to achieve medication optimization
   4. Collaborative, team-based care and privileging
   5. Professional development and maintenance of competence
      a. Board certification and recertification
      b. CPE
      c. Maintenance of licensure
      d. Participation in formal and informal development activities
   6. Professionalism and ethics – Demonstrate the traits of:
      a. Responsibility
      b. Commitment to excellence
      c. Respect for others
      d. Honesty and integrity
      e. Care and compassion for others
      f. High ethical standards
      g. Legal and regulatory compliance
   7. Research and scholarship
   8. Other
      a. Education and training
      b. Mentorship
      c. Management and leadership
      d. Policy and service development and implementation
      e. Consultation
      f. Performance/Quality improvement

B. Scope of Critical Care Pharmacy Services (Crit Care Med 2020;48:1375-82; Position Paper on Critical Care Pharmacy Services [a joint effort of SCCM, ACCP, and ASHP])
   1. The task force provided 82 recommendation statements across five distinct domains (see Table 1 for details):
      a. Patient Care (n=34)
      b. Quality Improvement (n=21)
      c. Research/Scholarship (n=9)
      d. Training/Education (n=10)
      e. Professional Development (n=8)
   2. Gradations of pharmacy practice
      a. Essential: These are practice recommendations that are considered vital to the provision of pharmacy care to ICU patients.
      b. Desirable: Offers recommendations that are more specialized and specific to the ICU beyond the essential recommendations
      c. Essential versus desirable recommendation statements are further categorized based on the level of ICU care being provided (see section V.D.1 in the text that follows for descriptions of level I, II, and III ICUs)

C. Critical Care Pharmacist as Educator: The critical care pharmacist has several educational missions and obligations (Pharmacotherapy 2011;31:135e-175e; Ann Pharmacother 2006;40:612-8; Pharmacotherapy 2002;22:1484-8; Crit Care Med 2000;28:3746-50 ; Crit Care Med 2020;48:1375-82), and teaching methods and techniques vary depending on intended audience and content. The clinical pharmacist must develop comfort and expertise with a wide range of teaching styles and techniques to be successful as an educator in the ICU setting.
   1. Pharmacy students and residents: Content has to be at a level appropriate to learners who may or may not have a primary interest in critical care. Active learning strategies must be incorporated with didactic approaches that are more traditional. For this audience, the clinical pharmacist has primary responsibility for assessment/grading.
      a. Clinical practice training
         i. Role modeling (I do, you watch)
         ii. Coaching (I do, you help … then … you do, I help)
         iii. Mentoring (You do, I watch)
         iv. Facilitating (you do, I monitor)
      b. Case-based teaching (point-of-care teaching)
      c. Hands-on demonstrations of equipment, technology, and devices used in the ICU
      d. Clinical conferences/topic discussions
      e. Assigned readings
      f. Journal club
      g. Quality improvement projects
      h. Writing assignments
         i. Case reports
         ii. Guideline/protocol development
         iii. Pharmacy and therapeutics (P&T) monographs
      i. Drug information response
      j. Medication use evaluations

2. Critical care team: More heavily focused on the specifics of critical care therapeutics. Content and sophistication will vary depending on audience (e.g., physicians vs. nurses). Audience is assumed to have a primary interest in critical care.
    a. Case-based, point-of-care teaching (bedside rounds)
    b. Didactic teaching
        i. Teaching rounds/conferences
        ii. In-service education
        iii. Grand rounds
        iv. Basic science lectures
    c. Critical care–specific journal club
    d. Collaboration on guidelines/protocols
    e. Quality improvement projects
3. Pharmacist colleagues: May not have a primary focus or interest in critical care. Content may be focused on specific pharmacotherapeutic issues (e.g., pharmacokinetic principles in the critically ill) that often arise during cross-coverage. May include pharmacists taking a nonconventional path to critical care practice.
    a. Didactic lectures (e.g., clinical conferences, topic-specific lectures)
    b. Hands-on demonstration of equipment, technology, and devices used in the ICU
    c. Case-based, point-of-care teaching
    d. Journal club
    e. Competency-based programs – Lead to credentialing according to demonstrated skills
4. Other trainees: Often includes a mix of backgrounds and interests (e.g., critical care fellows, anesthesia residents, medicine residents, emergency department (ED) residents, fourth-year medical students, nursing students, advanced-practice provider (APP) students, and dietary students). Content has to be appropriate for the predominant audience and baseline understanding of the topic.
    a. Didactic lectures
        i. Teaching rounds
        ii. Clinical conferences
    b. Point-of-care teaching – Bedside rounds
5. Patients and families: With an increasing emphasis on patient and family satisfaction and a greater involvement of the patient and family in shared decision-making, the need has increased for critical care pharmacists to be available as resources and provide education to patients and families about common medications and expected response in the ICU. Another emerging area of critical care pharmacist involvement is patient and family care and education in post-ICU clinics to address the challenges associated with post-ICU care for those who survive their ICU admission. The SCCM-sponsored THRIVE initiative promotes the growth of these interprofessional clinics.
    a. Techniques
        i. Simple language, basic content
        ii. Teach-back technique to assess understanding
        iii. Frequent reinforcement
        iv. Motivational interviewing techniques
        v. Open-ended questions to understand what is important to the patient and family
    b. Content
        i. Medications being initiated in the ICU
        ii. Why the medication is being used – Have goals different from home medications
        iii. What to expect – Effects, adverse effects, changes in patient interaction, etc.
        iv. Expected duration of new medications
        v. What factors are monitored to see whether medications are helping or hurting

D. Critical Care Services (Crit Care Med 2003;31:2677-83): ACCM recommended critical care services and personnel according to the level of care being provided. ICUs were defined as levels I, II, and III.
   1. Levels of ICU services
      a. Level I
         i. Comprehensive critical care for a wide variety of patient populations with a high level of specialization
         ii. Requires broad range of comprehensive support, including pharmacy services, respiratory therapy, clinical nutrition, pastoral care, and social services
         iii. Often fulfills an academic mission
      b. Level II
         i. Comprehensive critical care but may not provide care for certain patient populations
         ii. Must have transfer protocols in place for patients with special needs
         iii. Comprehensive support services must be available.
         iv. May or may not have an academic mission
      c. Level III
         i. Provides stabilization, but has limited ability to provide comprehensive critical care
         ii. Must have transfer protocols in place for patients requiring level I and II critical care services
         iii. Support services are often limited in scope.
   2. Critical care pharmacy services (level I and II ICUs)
      a. Reiterates pharmacist and pharmacy services defined in 2000 guideline (Crit Care Med 2000;28:3746-50)
      b. Emphasizes the importance of clinical pharmacists as required members of the patient care team
      c. Qualifications and competence of the critical care pharmacist in ICU therapeutics are defined as essential. Acknowledges several pathways, including advanced degrees, residency, fellowship, and other specialized practice experiences.
      d. ICUs with an academic mission should provide protected time for pharmacist participation in scholarly activities and appropriate knowledge and skills to provide education to critical care nurses, physician trainees, and physicians.
      e. Nonacademic centers should provide time for maintenance of competence and maintain current certification.

E. Principles and Values of Team-Based Health Care
   1. SCCM and ACCP have long promoted the team-based care model for critical care as a standard, including clinical pharmacists as essential staff (Crit Care Med 2006;34(3 suppl):S46-51).
   2. An Institute of Medicine (now National Academy of Medicine) discussion paper delineated the core principles and values of highly functioning interprofessional health care teams (see https://nam.edu/wp-content/uploads/2015/06/VSRT-Team-Based-Care-Principles-Values.pdf).
      a. Definition of team-based care: *Team-based health care is the provision of health services to individuals, families, and/or their communities by at least two health providers who work collaboratively with patients and their caregivers—to the extent preferred by each patient—to accomplish shared goals within and across settings to achieve coordinated, high-quality care.*
      b. Five personal values of effective members of high-functioning teams:
         i. Honesty: Includes effective, transparent communication. Essential to building mutual trust.
         ii. Discipline: Each team member carries out roles and responsibilities in a highly disciplined approach, even when inconvenient or difficult.
         iii. Creativity: Maintains excitement around addressing new and difficult challenges. Sees opportunity in both successes and failures.

iv. Humility: Equal respect of all members, regardless of level of training or role – Not tied to traditional hierarchical thinking in health care. Recognizes that all members of the team are susceptible to mistakes.
v. Curiosity: Dedicated to reflection and continuous improvement

c. Five principles of team-based health care
i. Shared goals: Clearly articulated, understood, and supported goals are established by the team that are consistent with the patient and family wishes. The patient and family are actively involved in establishing the goals of care as members of the team.
ii. Clear roles: Each team member's functions, responsibilities, and accountabilities are clearly established and understood by the team. Efficiency and logical division of labor are achieved. Although autonomy is important, flexibility of roles and collaboration exist as needed.
iii. Mutual trust: Establishing and maintaining trust, as well as openness to address questions about or breaches of trust, are essential. Mutual trust permits individual team members to function to their highest potential and rely on other team members to follow through on their commitments.
iv. Effective communication: Tightly linked to mutual trust. The team has consistent channels for candid and complete communication by all team members and in all situations.
v. Measurable processes and outcomes: The team develops and implements accurate and timely measures of successes and failures and uses the results to track and improve performance. Measures fall into two categories: Process/outcome measures and measures of team function. This principle is typically the most challenging for a team to implement and sustain effectively.

3. Critical care teams – Gap analysis: When considering the core principles of team-based care, critical care team members should evaluate their team structure and performance against these five principles. Effective teams are much more than patient care rounds by a mix of health care professionals. Common questions to consider when evaluating potential gaps should include:
a. Shared goals
i. Are the patient and family goals for critical care routinely incorporated into the care plan?
ii. Are the patient and family viewed as active members of the team during the establishment of goals?
iii. Are there clearly articulated and understood goals that are agreed on by all members of the team during the provision of care to all ICU patients and the work of the team in the care of that patient?
iv. Is progress toward the goals routinely reevaluated in light of the changing course and evolving perspective of the patient and family? Are goals adjusted or refined throughout the dynamic course of the critical care admission as needed?
v. Are there adequate organizational resources and commitments to permit effective establishment of shared goals in the treatment of ICU patients?

b. Clear roles
i. Are each team member's functions, responsibilities, and accountabilities clearly defined? Can each team member articulate and understand the role of the other team members?
ii. Are the roles and responsibilities of each team member focused on the shared goals of the team and patient?
iii. Is there clear respect for the contributions of each team member from a nonhierarchical, interdependent perspective?
iv. Is each team member introduced (and reintroduced) to the patient and family, including a lay description of each member's role and responsibility?
v. Does each team member go about his or her responsibilities with a reasonable degree of autonomy?
vi. Is there a clear team leader? Does the leadership role vary according to individual circumstances, problems, or environment?

vii. Does the team have a reasonable balance between autonomous functions and collaboration?
viii. Are there adequate organizational resources and commitments for professional development, team education, facilitation of communication, and restructuring of care processes to support the effective division of labor?

c. Mutual trust
   i. Does an environment of mutual trust and support exist among the ICU team? Can breaches of trust be openly discussed and addressed between team members without a detrimental impact on professional or personal relationships?
   ii. Does the hiring process include a focus on the personal and professional values that support an environment of mutual trust? Do members of the ICU team participate in the hiring process across traditional departmental siloes?
   iii. Is the team effective at establishing and maintaining mutual trust with patients and families? Are effective communication skills used to explain the process of establishing goals, sharing information on progress, and incorporating effective negotiation and conflict resolution skills?
   iv. Does the team regularly participate in non–patient care activities (e.g., social interaction) that allow team members to develop greater trust and know each other at many levels?
   v. Is there adequate organizational support of the elements necessary to establish mutual trust among teams?

d. Effective communication
   i. Has the team established a high priority for open, direct, clear, consistent, professional communication between team members?
   ii. Does communication take advantage of all potential modes and technologies of communication for efficiency and convenience?
   iii. Do members of the team use effective listening skills, recognizing that deep listening to the input of all team members, including patients and families, is an essential component of effective communication?
   iv. Are signs of tension and unspoken conflict in the communication process regularly recognized and addressed to improve team communication skills and effectiveness?
   v. Does effective communication occur across the team regardless of traditional hierarchical structures in health care?
   vi. Are the organizational elements for effective communication available to the team?

e. Measurable processes and outcomes
   i. Has the team identified and implemented reliable, timely, and ongoing measures of team performance?
   ii. Are these measures focused on both process/outcomes of care provision and team function or effectiveness?
   iii. Are measures of patient and family satisfaction included in the assessment process?
   iv. Are measures of team member satisfaction included in the team assessment?
   v. Does the team regularly report its measures of success and failure, both internally to the team and to others in the organization?
   vi. Are performance data regularly used for process improvement with respect to both patient care and team function?
   vii. Does the team use any standardized tools to assess team function and quality?
   viii. Are organizational resources and commitment adequate to permit teams to adequately measure quality of patient care and team function?

F. Other Standards
   1. The Joint Commission
      a. Medication management chapter
         i. High-alert, hazardous medication standards
         ii. Look-alike/sound-alike medications
         iii. Monitoring of medication response
         iv. Adverse drug event detection, evaluation, and reporting
         v. Duplicate therapy
      b. National Patient Safety Goals
         i. Two-factor patient identification
         ii. Medication reconciliation
         iii. Safe medication use and labeling
         iv. Anticoagulation management and education
         v. Antimicrobial stewardship program
   2. Centers for Medicare & Medicaid Services (CMS) conditions for participation (42 CFR 482)
      a. Quality assurance and performance improvement programs (§482.21)
         i. Medical errors
         ii. Adverse events
      b. Preparation and administration of medications (§482.23)
      c. Medical records requirements (§482.24)
      d. Pharmaceutical services (§482.25)
         i. Policies and procedures to minimize drug errors
         ii. Adverse drug reaction and medication error detection and reporting
         iii. Drug information standards

VI. TRAINING OF CRITICAL CARE PHARMACISTS

A. Positions and Policy
   1. ACCP position
      a. ACCP clarified its position concerning qualifications of clinical pharmacists providing direct patient care in a 2013 Board of Regents commentary (Pharmacotherapy 2013;33:888-91): Clinical pharmacists providing direct patient care "should possess the education, training, and experience necessary to function effectively, efficiently, and responsibly in this role. Therefore, ACCP believes that clinical pharmacists engaged in direct patient care should be board certified (or board eligible if a Board of Pharmacy Specialties [BPS] certification does not exist in their area of practice) and have established a valid collaborative drug therapy management (CDTM) agreement or have been formally granted clinical privileges by the medical staff or credentialing system within the health care environment in which they practice."
      b. Board certification
         i. ACCP considers BPS certification the cornerstone of eligibility for direct patient care.
         ii. Eligibility
            (a) Graduate of an accredited school of pharmacy
            (b) Pharmacy licensure
            (c) Postgraduate residency training in area of specialization (i.e., PGY2 pharmacy residency), or PGY1 pharmacy residency and at least 2 years of experience in specialty area with at least 50% of time in the BPS scope of specialty area, or at least 3–4 years of relevant experience with at least 50% of time practicing in the specialty area

iii. ACCP has expressed that postgraduate residency training is the preferred training pathway for clinical pharmacists providing direct patient care in previous position statements and a white paper.

2. ASHP policy
   a. Policy 2027: "Pharmacists who provide direct patient care should have completed an ASHP-accredited residency or have attained comparable skills through practice experience."
   b. ASHP has no policy directly related to the provision of direct patient care in a specialty practice area.

B. Potential Workforce Demands
   1. Hospital data
      a. 6129 hospitals in the United States – 919,649 staffed beds (2023 American Hospital Association survey data; www.aha.org/system/files/media/file/2023/05/Fast-Facts-on-US-Hospitals-2023.pdf)
      b. According to 2018 AHA survey data, there were 107,276 adult, pediatric, and neonatal ICU beds in community hospitals across the United States (www.aha.org/statistics/fast-facts-us-hospitals).
   2. Critical care pharmacists
      a. No accurate database to indicate the number of pharmacists spending 50% or more of time in critical care
      b. Direct patient care by pharmacist provided in 62.2% of ICUs in the United States in 2006. This represents primarily fundamental-level services (Ann Pharmacother 2006;40:612-8). A more recent survey of 1220 institutions revealed 70.8% of ICUs had direct clinical ICU pharmacy services, which is a greater than 13% increase compared with 2006 (Crit Care Explorations 2021;3:1-11).
      c. A recent survey of 185 critical care pharmacists indicated that the pharmacist/patient ratio is variable, with 84% of respondents caring for more than 15 patients. In addition, 30% of respondents expressed concern about safety relative to workload, and 50% indicated they felt overworked. This survey suggests that ICUs with pharmacist coverage may be understaffed (JACCP 2020;3:68-74).
      d. A review of ACCP, SCCM, ASHP, and American Pharmacists Association (APhA) membership records identified 2928 individual pharmacists indicating specialization in critical care at the time of the 2012 petition for recognition of critical care as a specialty (https://www.accp.com/docs/positions/petitions/Final_CRITICAL_CARE_PETITION_For_BPS_Post.pdf).
      e. A survey sent to those 2928 pharmacists yielded 504 responses (https://www.accp.com/docs/positions/petitions/Final_CRITICAL_CARE_PETITION_For_BPS_Post.pdf).
         i. Out of the responses, 476 reported that they practiced in critical care (94%).
         ii. Of those 476, 91% responded that their practice met the definition of critical care pharmacy as a specialty.
         iii. Among the respondents, 74% indicated they spent at least 50% of their time practicing in the ICU.
         iv. More than 80% of respondents completed residency or fellowship training in critical care.
      f. A survey of employers of ICU pharmacists yielded 204 responses (https://www.accp.com/docs/positions/petitions/Final_CRITICAL_CARE_PETITION_For_BPS_Post.pdf).
         i. Collectively employed 1034 full-time equivalent (FTE) critical care pharmacists
         ii. Recruited 256 critical care pharmacists during the previous 3 years
         iii. Estimated a need to hire 234–243 critical care pharmacists in the next 3 years
         iv. Of the respondents, 99.5% estimated the demand for critical care pharmacists to grow or remain stable at their site during the next 5 years.
      g. Critical Care Societies Collaborative (CCSC) – http://ccsconline.org/workforce
         i. Collaborative effort of several stakeholder organizations in critical care to define the workforce shortage in critical care and advocate for federal action to address the problem

ii. Most of this work has focused on intensivist and ICU nurse shortages, but there is also recognition of shortages of other professionals, including critical care pharmacists.

h. Current and objective quantification of critical care pharmacist shortage or demand is unavailable.

C. Training Recommendations and Capacity

1. Minimum requirements for all levels of ICU service (levels I–III)
   a. Graduate of Accreditation Council for Pharmacy Education (ACPE)-accredited school or college of pharmacy
   b. Licensure and registration by a state board of pharmacy
2. Conventional or preferred postgraduate training
   a. PGY1 pharmacy practice residency based in a hospital
   b. PGY2 critical care residency or fellowship
      i. First critical care pharmacy residency described: 1981 (The Ohio State University)
      ii. ASHP critical care pharmacy residency standards published in 1990
      iii. 185 ASHP-accredited critical care residencies in 2023; increased from 8 in 2001 and 39 in 2005
      iv. Most PGY2 critical care residents are somewhat or very satisfied (91% and 76%, respectively) with their program and mentorship according to a 2012 survey.
      v. Critical care pharmacy research training: Long history of fellowship training; however, the *ACCP Directory of Residencies, Fellowships, and Graduate Programs* lists fellowship programs with a primary or secondary focus on critical care
3. Nontraditional alternative paths: There is no widely accepted or clearly defined alternative pathway to specialty experience and competence in critical care pharmacy. Some potential pathways and components of a self-directed training program are outlined in the text that follows. The extent and variety of experiences needed may be determined by the practice setting, level of care to be provided, baseline knowledge, availability and willingness of qualified mentors, and other personal and professional skills of the individual. Although many potential paths are defined later, those that provide continued, practical experience during a prolonged period in a supervised or mentored environment are considered of greatest value in developing competency in the ICU setting.
   a. Mentored or supervised clinical practice experience without residency
      i. Clinical practice experience must be hands-on and team based under supervision.
      ii. Mentors may be PGY2- or fellowship-trained critical care pharmacists, clinical pharmacists with equivalent experience, critical care faculty from affiliated schools of pharmacy, intensivist physicians, and/or other critical care professionals.
      iii. Several mentors may best meet the variety of needs of the mentee pharmacist.
      iv. Reinforced by frequent reading and analysis of the critical care primary and secondary literature, journal club participation, and frequent critical discussions of the clinical implications of the primary literature
      v. Normally, expect at least 3–4 years of mentored/supervised experience to gain competency for independent clinical practice (optimal services) in level I and II ICUs. Shorter periods may be adequate to provide lower levels of service to level II and III ICUs.
   b. PGY1 with supervised/mentored ICU clinical practice experience
      i. Mentored clinical experiences similar to those described earlier
      ii. PGY1 with critical care experiences during residency may be adequate to provide fundamental and desirable services to level II and III ICUs.
      iii. Normally, expect 2–3 years of mentored/supervised experience to gain competency for independent clinical practice (optimal services) in level I and II ICUs.
   c. ASHP critical care traineeship (https://www.ashpfoundation.org/leadership-development/traineeships)
      i. Previously offered through the ASHP Foundation

ii. Four-month distance education component – Independent reading, web-based education, and teleconference case studies
iii. Two-week on-site experiential training
iv. Post-experiential training activities
v. Is not a comprehensive training program but can be a valuable component of a training program for nontraditional-path clinical pharmacists

d. Other potential components of a nonconventional training program: Actual program structure will vary depending on the available resources, practice environment, baseline knowledge and skills of the pharmacist, and institutional support.
i. Graduate degree (e.g., master's degree)
ii. Continuing education (CE) programming – Live, web based, print
iii. Attendance at national and regional critical care meetings – CE, networking, research presentations
iv. Fundamental Critical Care Support course completion (www.sccm.org/Fundamentals)
v. ACLS, advanced trauma life support (ATLS), and/or pediatric advanced life support (PALS) training and certification
vi. Regular participation in the SCCM Clinical Pharmacy and Pharmacology Section national journal club
vii. SCCM Clinical Pharmacy and Pharmacology Section mentor program – Long-distance mentoring program
viii. Self-arranged experiential rotations at peer institutions under the supervision of a qualified critical care pharmacist
ix. Visiting professor or scholar programs to bring specialized expertise to the clinical site for on-site experiential training and didactic teaching
x. Policy, guideline, and protocol development for critical care pharmacotherapy–related issues under the supervision of qualified peers
xi. Critical care pharmacy service or program development, implementation, and outcome measurement under the supervision of qualified peers

D. Post-training Professional Mentoring and Critical Care Pharmacy Training (Am J Pharm Educ 2013;77:1-7; Am J Pharm Educ 2003;67:1-7)
1. Mentor-protégé relationship
a. Symbiotic, nurturing relationship between two adults
b. Assist each other in meeting shared career objectives
c. Attributes of a successful mentor-protégé relationship (see Box 1)
d. Mentor typically 15–20 years older than protégé
2. Mentor should fulfill five functions:
a. Teaching – New knowledge, skills, and attitudes
b. Sponsoring – Helps protégé reach career goals, assists in networking, vouches for abilities, offers protection from threats
c. Encouraging – Affirming, challenging, inspiring
d. Counseling – Listening, probing, advising during difficult challenges
e. Befriending – Acceptance, understanding, and trust

Box 1. Attributes of Successful Mentor-Protégé Relationships

Mentor Qualities
Strong interpersonal skills Technical competence/expertise Knowledge of organization and profession Status/prestige within the organization and profession Willingness to be responsible for someone else's growth and development Ability to share credit Patience
Protégé Qualities
Self-perceived growth needs A record of seeking/accepting challenging assignments Receptivity to feedback and coaching Willingness to assume responsibility for own growth and development Ability to perform in more than one skill area
Relationship Qualities
Voluntary Mutual benefits perceived and derived from the relationship No conflicts of interest/competition between mentor and protégé Not confined to professional or business interests

Adapted from: Haines ST. The mentor-protégé relationship. Am J Pharm Educ 2003;67:Article 82, p 2.

3. Phases of the mentor-protégé relationship:
    a. Initiation phase
        i. Weeks to months in duration
        ii. Begin work together
        iii. Mentor coaches protégé, and protégé may provide technical assistance.
    b. Cultivation phase
        i. 2–5 years in duration
        ii. Both individuals realize personal and professional benefits.
        iii. Deeply intimate and personal bonds are formed.
    c. Separation phase
        i. Typically months in duration
        ii. Protégé no longer requires guidance and begins to seek more autonomy.
        iii. Mentor may think they have been deserted, whereas protégé may believe they are being held back.
        iv. Resentment or hostility may lead to end of relationship.
    d. Transformation phase
        i. Years in duration (lifelong)
        ii. Peer relationship evolves.
        iii. Mutual sense of gratitude and appreciation
4. Voluntary versus arranged relationships
    a. Increasingly, organizations are establishing mentoring programs with assigned mentors.
    b. Successful mentoring relationships are voluntary and based on mutual respect.
    c. Successful and powerful people are not necessarily good mentors.
    d. The factors that lead to mentor-protégé relationships are unclear and may be difficult to create through assignment of mentors.

e. Factors that contribute to successful mentorship:
   i. Common interests
   ii. Common purpose
   iii. Desire on the part of the mentor to participate
   iv. Mentor and protégé must be able to spend time together.
   v. Persistent and regular interaction between mentor and protégé
f. Formal mentoring programs can be successful, but less so than voluntary relationships.

5. Mentoring and critical care training
   a. Beyond formal residency/fellowship programs, mentor-protégé relationships are essential to the formal development of critical care pharmacists.
   b. Developing critical care pharmacists should seek out mentors with similar interests and purpose who can help them fill gaps in their knowledge, skills, and attitudes relative to critical care practice.
   c. Over time, critical care pharmacists may have several mentor-protégé relationships to meet evolving educational and experiential needs.
   d. Experienced and successful critical care pharmacists should volunteer to mentor junior pharmacists, residents, and students and take their roles as mentors seriously by being kind, helpful, supportive, and encouraging.
   e. SCCM Clinical Pharmacy and Pharmacology Section mentoring services: Practice, education, administration, scholarship

VII. CONTINUING PROFESSIONAL DEVELOPMENT (https://www.acpe-accredit.org/continuing-professional-development/; Resident Survival Guide. Lenexa, KS: ACCP, 2011:123-37; Am J Health Syst Pharm 2004;61:2069-76)

A. General Considerations
   1. Lifelong learning by health care professionals is both a necessity and an obligation to several stakeholders.
   2. CPD is a multifaceted, self-directed, holistic, outcomes-focused approach to lifelong learning.
   3. CPD is a career-long iterative process with continuous cycles, rather than a start and a finish.
   4. Sustained career growth and success are more dependent on CPD than on early career education and training.
   5. CPD should be closely integrated into daily practice and the work environment for success and sustainability.

B. Stakeholders in CPD: Stakeholders may have a role in contributing to lifelong learning, benefiting from the sustained competency of the clinical pharmacist, or both.
   1. Pharmacist-learner (self)
      a. Most at stake
      b. Primarily responsible for developing a self-directed, structured approach to learning and assessment
      c. Must develop an approach that is flexible, integrated, and capable of being sustained throughout decades of practice
      d. Must be prepared to commit personal time to CPD
   2. Employer
      a. Has both an obligation to and an expectation of the clinical pharmacist relative to CPD
      b. Provision of resources
         i. Travel funding (varies depending on institutional budget)
         ii. Access to electronic databases and literature

iii. Environment that promotes sharing and learning (clinical conferences, journal club, open discussion and debate among colleagues, etc.)
iv. Protected time to pursue educational opportunities

c. Establish a credentialing and privileging process that incorporates CPD expectations.
d. Aligning personal development goals with institutional priorities is mutually beneficial and may increase employer support.
e. Employer benefits from sustained and expanded competencies of clinical pharmacist and should incorporate into hiring, retention, and promotion decisions

3. Colleagues
   a. Contribute to lifelong learning of the clinical pharmacist
      i. Case-based discussion and debate on daily rounds
      ii. Drug-related questions
      iii. Interdisciplinary teaching rounds
      iv. Clinical conferences, journal clubs
      v. Inclusion in collaborative scholarly activities
   b. Benefit from lifelong learning of the clinical pharmacist
      i. Greater quality and sophistication of contributions to team-based care of critically ill patients
      ii. Educational offerings by the clinical pharmacist
      iii. Collaboration around scholarly activities
      iv. ICU-related treatment guidelines and protocols developed by or in collaboration with the clinical pharmacist
4. Students, residents, and fellows
   a. Contribute to lifelong learning of the clinical pharmacist
      i. Assisting in identifying gaps in their own knowledge
      ii. Creating incentive to maintain competency through CPD
      iii. Regularly challenging applicability and relevance of professional knowledge and skills
   b. Benefit from current, relevant knowledge and skills being incorporated into:
      i. Teaching
      ii. Role modeling/coaching
      iii. Mentoring
   c. CPD is a lifelong obligation of pharmacists who accept responsibility for training future clinical pharmacists.
   d. The best trainees seek out the most competent teachers, preceptors, and mentors.
5. Patients
   a. Greatest beneficiary of clinical pharmacist CPD
   b. Providing the best possible care to ICU patients should be the biggest motivator for the clinical pharmacist to pursue CPD.
   c. Well-informed patients will seek out the most competent and capable health care professionals.

C. CPD Process: The CPD process is structured around four essential steps. A potential fifth step is documentation of the process, but that should be an integral part of each step, not a separate process.
   1. Reflection
      a. Self-assessment process
      b. Evaluation and feedback from others
         i. Coworkers
         ii. Colleagues
         iii. Employer
         iv. Patients
         v. Pharmacy trainees (students/residents)
         vi. External peer review (e.g., organizational fellowship status; adjunct faculty appointment review)

   c. Personal SWOT (strengths, weaknesses, opportunities, and threats)
      i. Assessment of internal strengths and weaknesses related to knowledge, skills, experiences, and behaviors
      ii. Assessment of external environmental factors for opportunities and threats
      iii. Goal is to identify learning needs and opportunities that exist to address those needs.
   d. Reflection should be both scheduled and episodic.
      i. Annual performance evaluation/self-evaluation (scheduled)
      ii. Some set or chosen anniversary date (scheduled)
      iii. After the care of a complex or difficult patient (episodic)
      iv. After an interaction with a challenging student or resident (episodic)
   e. Result of reflection is to identify two or three specific, well-defined, and achievable learning needs.
2. Plan
   a. Develop a personal development plan (PDP) to address the needs and opportunities identified during reflection.
   b. Includes learning objectives that are specific, measurable, achievable, relevant, and timed (SMART)
   c. Identifies resources needed to address the PDP
   d. Evaluates the availability and access to needed resources and modifies plan accordingly
   e. The PDP should be regularly reassessed and adjusted as needed.
3. Act
   a. Develop an action plan to implement the PDP.
   b. The action plan will need to incorporate a variety of learning strategies and methods (see text that follows).
   c. Incorporating the action plan into the daily practice activities is key to success and sustainability. CPD should not be considered an additional burden.
4. Evaluate
   a. Evaluate the effectiveness of the action plan for achieving the learning objectives of the PDP.
      i. Did the activities provide adequate content, depth, and hands-on experiences to truly address the learning objectives and meet the needs identified during reflection?
      ii. Did the activities stay focused on the learning objectives, and were timelines adhered to adequately?
      iii. Were all competencies adequately addressed?
      iv. How did the CPD activities affect the pharmacist-learner and possibly the patient (often very challenging to measure)?
   b. Evaluation is expected to lead to the next round of reflection and restart the continuous and iterative process of CPD.
5. Portfolio
   a. Process of documenting the CPD progression
   b. Although the format may be standardized by employer, regulatory authorities, CE providers, or others, the content should be individualized to reflect the needs, actions, and assessments of the pharmacist-learner.
   c. Is a dynamic, living document that reflects the continuous, iterative nature of CPD
   d. Examples of CPD portfolio formats/templates:
      i. https://www.acpe-accredit.org/pdf/CPD_Portfolio.pdf
      ii. North Carolina CPD Learning Portfolio, North Carolina Board of Pharmacy, https://bpsweb.org/2023/09/29/introduction-to-cpd/
      iii. www.ocpinfo.com/practice-education/qa-program/learning-portfolio/

D. CPD Learning Strategies and Methods
   1. CPE
      a. CPD is not a replacement for CPE, but CPE should be one component of a PDP.
      b. ICU pharmacists should focus on CPE programming that meets their defined educational needs and incorporates several different techniques that will help meet the full range of competencies needed for clinical practice in the ICU.
      c. Accreditation standards maintain minimum quality assurance of CPE activities.
      d. CPE credits are the most widely used "currency" by regulatory bodies, accrediting agencies, and other organizations as a proxy for professional competency, and this is unlikely to change in the near or intermediate term.
      e. Traditional didactic lecture-style CPE activities have several limitations toward achieving CPD learning objectives.
         i. Often non-curricular
         ii. Limited influence on changing practice
         iii. Educational outcomes may not align with the individual's needs.
         iv. Content is sponsor or speaker driven.
         v. Opportunity for bias (or perception of bias), depending on source of support
         vi. CE efforts are often fragmented across professions (not interdisciplinary).
      f. CPE providers are expanding the diversity of educational methodologies and techniques to include interaction, experiential learning, simulation, discussion and debate, and role-playing, among others.
      g. Limited evidence suggests that live CE over print, multimedia format, and a series of programs on a curricular theme is the most effective CE method.
   2. Short courses or seminars
      a. Certificate or credentialing programs
         i. ACLS
         ii. ATLS
         iii. PALS
         iv. Emergency neurological life support (ENLS)
         v. Teaching certificate programs
      b. Structured curricular programs
         i. ACCP Academies
         ii. Fundamental Critical Care Support (FCCS) and Pediatrics Fundamental Critical Care Support (PFCCS) course
         iii. Pharmacotherapy of Neurocritical Care Series (PONS) offered by the Neurocritical Care Society
   3. Membership and participation in national organizations
      a. ACCP; Critical Care PRN
      b. SCCM; Clinical Pharmacy and Pharmacology Section
      c. ASHP; Section of Clinical Specialists and Scientists
      d. Several specialty organizations related to critical care (American College of Chest Physicians, American Trauma Society, Neurocritical Care Society, American Burn Association, etc.)
   4. Primary and secondary literature
      a. Reading, analyzing, and applying the relevant literature should be central to any strategy of professional development.
      b. No gold standard strategy for staying current with the literature

c. Many "foraging" strategies will need to be considered and used.
   i. Review table of contents of high-impact journals in critical care (e-mail or rich site summary [RSS] push technology) (e.g., *Critical Care Medicine*, *Intensive Care Medicine*, *Chest*, *American Journal of Respiratory and Critical Care Medicine*, *Journal of Trauma and Acute Care Surgery*, *Journal of Critical Care*).
   ii. Topic alerts (e-mail or RSS) for critical care articles from high-impact multispecialty journals (e.g., *New England Journal of Medicine*, *Annals of Internal Medicine*, *JAMA*, *The BMJ*, *Lancet*)
   iii. Scan high-impact pharmacy specialty journals for critical care articles (e.g., *Pharmacotherapy*, *Annals of Pharmacotherapy*, *American Journal of Health-System Pharmacy*).
   iv. Use of saved search strategies with automatic e-mail alerts on a scheduled interval (e.g., PubMed, PubCrawler, Ovid Medline)
   v. Subscribe to a medical information alert service with high and transparent standards for validity, relevance, and contextual interpretation of the data (e.g., Essential Evidence Plus, FPIN [Family Physicians Inquiries Network] Clinical Inquiries, BMJ Clinical Evidence, Cochrane for Clinicians).
   vi. Scan review journals relevant to critical care (e.g., *Critical Care Clinics*).
   vii. Identify high-quality, relevant, and contemporary clinical practice guidelines for critical care therapeutics (e.g., National Guideline Clearinghouse, PubMed Clinical Queries, MD Consult).
   viii. Use up-to-date systematic reviews (e.g., *Cochrane Database of Systematic Reviews*, *Agency for Healthcare Research and Quality [AHRQ] Evidence-Based Practice Center Evidence-Based Reports*).
   ix. Selective use of other resources (e.g., evidence-based summaries such as Bandolier, Clinical Evidence), critically appraised topics, point-of-care review services (e.g., UpToDate, Medscape), SCCM CPP Section Pharmacotherapy Literature Updates, and meta-search engines (e.g., Trip database)

d. The tools and resources available for staying current with the literature is a rapidly evolving, dynamic market. The individual pharmacist will need to stay current to maximize use of the literature and will need to adapt his or her strategy over time.

5. Discussion and debate with colleagues, mentors, and other content experts
   a. Therapeutic dilemmas
   b. Complex cases
   c. Primary literature
   d. Guidelines
6. Journal clubs/clinical conferences
7. Interdisciplinary, patient care rounds – Daily interactive discussions of diagnostics, disease states, therapeutics, monitoring, technology in the ICU, ethics, communication with patients and families, etc.
8. Guideline and protocol development for the ICU
   a. Translation of evidence to best practices
   b. Benchmarking with peer institutions
   c. Consensus building
   d. Project management – Implementation and measurement of outcomes
9. Point-of-care learning
   a. Refers to day-to-day learning opportunities
   b. Uncommon disease state or unexpected adverse drug reaction prompts reading and learning.
   c. Complex drug information questions from colleagues

VIII. CORE KNOWLEDGE BASE AREAS FOR PHARMACISTS CARING FOR CRITICALLY ILL PATIENTS (2024 BOARD OF PHARMACY SPECIALTIES CRITICAL CARE CONTENT OUTLINE) (SEE BOX 2)

Box 2. 2024 Board of Pharmacy Specialties Critical Care Content Outline

1	Critical Care
1A	**Critical Illness**
1A1	Pathophysiology
1A2	Critical Care Disease States
1A3	Chronic Disease States
1A4	Risk Factors for Critical Illness
1A5	Post-ICU Syndrome (PICS)
1A6	Medical Emergencies
1B	Medical Therapies and Devices
1B1	Mechanical Circulatory Support
1B2	Mechanical/Assisted Ventilation
1B3	Extracorporeal Membrane Oxygenation (ECMO)
1B4	Renal Replacement Therapies
1B5	Lines, Tubes, and Drains
2	**Therapeutics and Patient Management**
2A	**Treatment Planning**
2A1	Diagnosis and Diagnostic Testing
2A2	Pharmacodynamics and Pharmacokinetics
2A3	Goals of Care
2A4	Perioperative Care
2A5	Transitions of Care (e.g., medication reconciliation, disposition, stepdown)
2A6	Medication Delivery (e.g., routes of administration, delivery devices, anatomic alterations)
2A7	Pharmacoeconomics
2A8	Social Determinants of Health
2B	**Pharmacotherapy**
2B1	Pulmonology
2B2	Cardiology
2B3	Nephrology
2B4	Neurology
2B5	Gastroenterology
2B6	Hepatology
2B7	Dermatology
2B8	Immunology
2B9	Endocrinology
2B10	Hematology
2B11	Toxicology
2B12	Infectious Diseases
2B13	Supportive Care
2B14	End-of-Life Care
2B15	Fluids, Electrolytes, and Nutrition
2B16	Trauma and Burns

Box 2. 2024 Board of Pharmacy Specialties Critical Care Content Outline (*con't*)

2	**Therapeutics and Patient Management**
2C	**Treatment Outcomes and Monitoring**
2C1	Therapeutic Targets and Endpoints
2C2	Therapeutic Drug Monitoring
2C3	Physiologic Response Monitoring
2C4	Drug Interactions and Adverse Reactions
2C5	Patient and Caregiver Education and Counseling
3	**Professional Practice**
3A	**Quality of Care**
3A1	Medication Safety (e.g., REMS, VAERS, barcodes)
3A2	Risk Mitigation (e.g., failure mode effects analysis, root cause analysis)
3A3	Healthcare Regulations and Standards (e.g., FDA, TJC, CMS, NIOSH, USP<800>)
3A4	Quality Management/Improvement
3A5	Formulary Management
3A6	Clinical Decision Support Systems
3A7	Development and Justification of Pharmacy Services
3B	Practice Management
3B1	Practice Standards and Guidelines
3B2	Scholarship (e.g., statistics, research methods, literature evaluation, publications)
3B3	Human Subjects Research (e.g., IRB/IEC, compassionate use, right-to-try)
3B4	Interprofessional Healthcare Collaboration
3B5	Pharmacist Preceptorship and Mentorship
3B6	Alternative Healthcare Delivery Modalities (e.g., tele-ICU, post-ICU clinic)
3B7	Disaster Healthcare Management

Information from: Board of Pharmacy Specialties (BPS). Examination Specifications: Critical Care Pharmacy. Available at https://www.bpsweb.org/wp-content/uploads/BCCCP-Examination-Specifications-FALL-2023.pdf.

IX. DISSEMINATION OF CRITICAL CARE KNOWLEDGE

A. Reasons to Disseminate Knowledge
   1. Recognition by peers
   2. Promotion and tenure
   3. Ethical obligation of research
   4. Grantsmanship success
   5. Giving back to the discipline – Critical care pharmacy is a new and evolving specialty.
   6. Travel support

B. Venues for Disseminating Knowledge
   1. Peer-reviewed publications (see text that follows for greater detail)
      a. Traditional, print journals
      b. Open-access (electronic) journals
   2. Non–peer-reviewed publications
      a. Textbook chapter
      b. Pre-print publications
      c. Commentary/editorial
      d. Newsletter

e. Guideline
f. Compendia
g. CE material
h. Blogs

3. Abstract (see text that follows for greater detail)
   a. Poster
   b. Platform
   c. Regional, national, international meetings
   d. Virtual poster sessions
4. Presentation
   a. CE
      i. Live/lecture – Local, regional, national, international venues
      ii. Webinar
      iii. Recorded/archived
   b. Seminar or conference

C. Publication and Peer-Review Process
   1. Categories of publications (will vary by journal)
      a. Original research
      b. Systematic review (e.g., meta-analysis)
      c. Expert review
      d. Brief reports (e.g., preliminary or pilot data)
      e. Case reports
      f. Practice or educational insights (typically must include assessment of outcomes)
   2. Selecting a target journal
      a. Quality and importance of the publication
         i. First-tier journals – Highly important, innovative, and/or high quality
         ii. Second- and third-tier journals – To be considered when paper unlikely to be accepted in, or rejected by, first-tier journal
      b. Target audience and scope of the journal relative to content of publication – Looking for good match on both
      c. Seek input from coauthors, peers, colleagues concerning appropriate journal to target
      d. Publication fees (if applicable)
   3. Preparing the manuscript
      a. Comply with journal requirements.
         i. Historically published with first issue of each volume
         ii. Today, easiest to access online
         iii. Manuscript format
         iv. Margins, font, type size
         v. Abstract format
         vi. Word limits
         vii. Reference style and limits
         viii. Figures and tables
      b. Succinct, focused, non-repetitive – Economy of words
      c. Common weaknesses to avoid (original research) – In manuscript preparation, not underlying research
         i. Abstract does not match body of manuscript
         ii. Introduction fails to sell the importance and relevance of the objective(s)
         iii. Poorly worded or unclear study objective(s)
         iv. Methods without results; results without methods

v. Unnecessary duplication of results in tables and body of manuscript
vi. Rambling, unfocused discussion
vii. Failure to adequately address weaknesses of the study (all studies have them)
viii. Conclusions that reach beyond the data
ix. Several tables that can be consolidated
x. Unneeded figures (usually, simplistic presentations of data that can be presented parenthetically)
xi. Failure to cite the literature correctly or according to journal's requirements
xii. Exceeding word count limits – Both in abstract and in manuscript

4. Using citation manager software (EndNote, Reference Manager, RefWorks, etc.)
   a. May contain templates consistent with many biomedical journal requirements
   b. Actively cite the literature while writing
   c. Automatically re-sort the references during revisions
   d. Direct download of citations during literature searches
   e. Can include PDF files and your notes in citation file
   f. Develop libraries of commonly used citations
   g. Overall, can ease the writing and formatting process for publication
5. Submitting the manuscript
   a. Greatly simplified by web-based submission
   b. Follow download instructions carefully.
   c. Cover letter
      i. Communication to the editor
      ii. Declare category of publication (though now part of submission template).
      iii. Indicate corresponding author (also part of template).
      iv. Some journals encourage a brief explanation of why paper is being submitted to the journal – Relevance, importance, target audience. However, this is declining.
   d. Copyright release
      i. Electronic methods are increasingly used.
      ii. Each author must sign/submit.
      iii. Provide assurance that part or all of content has not been previously published and is not currently under consideration by another publisher. Usually excludes abstracts.
6. Conflicts of interest: All authors must provide conflict-of-interest statements.
7. Review and revision process
   a. Editorial review
      i. The editor or a member of the editorial board may review initially.
      ii. Looking for relevance to journal, general quality of the manuscript, composition, and readability
      iii. Failure to get past the editor's desk results in rejection
   b. Peer or scientific review
      i. Sent to peers with content expertise for review and critique
      ii. Typically, sent to two to five reviewers (varies by journal and internal criteria)
      iii. Most journals request a review to be returned in 10–30 days.
      iv. Reviewers are asked to focus on the quality of the research or content and importance (including relevance to the journal's audience), not copyediting details.
      v. Typical recommendation categories are as follows: Accept, minor revision, major revision, or reject.
      vi. Reviewers provide detailed comments and critique to be shared with the authors, and, often, comments to the editor that are not shared with the authors.

c. Editor response
    i. The editor (or designee) uses reviewer input to formulate a response to the authors.
    ii. If minor or major revision is requested, details of the required revisions are provided, often including the reviewers' specific comments. The editor often adds requests for revision.
    iii. A timeline for response is included. If manuscript is not resubmitted by deadline, opportunity to publish is usually surrendered.
    iv. A rejection decision is usually final.
d. Revision process
    i. The authors need to respond to each request. The authors need not agree with each request, but each must be responded to and defended if not revised as advised.
    ii. Common format is a letter to the editor restating each request, with the response to the request immediately following.
    iii. A manuscript incorporating all revisions is submitted with the letter. Some journals request a "track changes" version of the manuscript to ease the rereview process.
    iv. All authors must review the revisions and indicate their agreement with all changes.
    v. Revised manuscripts are often returned to the original peer reviewers for a second review, especially with major revisions. That may result in another round of revision and review.

8. After acceptance
    a. At some point (timelines vary by journal), the corresponding author will receive the galley proof. This is a copyedited, typeset version of the paper that will look like the final publication.
    b. The galley proof will come with comments that must be addressed.
    c. Deadline for galley submission may be as short as 48 hours.
    d. The galley proof must be read very carefully and compared with the manuscript to ensure that copyediting changes do not affect the meaning, tables are formatted as intended, figures and legends are correct, and references are in the correct order and format (there are often errors with references – better with electronic confirmation).
    e. Ideally, all authors should review the galley proof; however, that may not be practical. All authors should approve a review by the corresponding author.
9. Timelines (highly variable by journal)
    a. Important competitive metric for biomedical journals
    b. From submission to response – Can be 2–4 months
    c. Reply to decision – Can be 1 week to 3 months (depends on whether rereview occurs)
    d. Decision to publication – Can be an additional 3–6 months
    e. Sometimes an important consideration when selecting a journal – Manuscript can be tied up for long periods.
    f. Open-access and e-journals have a speed advantage – May have fees.

D. Abstracts and Scientific Presentations
1. Selecting a meeting
    a. Quality and relevance of presentation
    b. Prestige of the meeting relative to authors' career goals
    c. Membership and desire to support an organization
    d. Availability of travel funding
    e. Priorities of key coauthors
    f. Location of the meeting (unfortunate, but true …)
    g. Encore presentations permitted

2. Developing the abstract
    a. Succinct and effective
        i. Greatest impact in the least space – No unnecessary words
        ii. Use of identifiable abbreviations, but not to excess
        iii. If allowed, use of tables to present results (many disallow)
    b. Title must be brief, be on point, and capture the reader.
    c. Clearly stated purpose/objective (minimize introductory material). May need to limit to primary objective.
    d. Methods and analysis are concise but of adequate detail to permit review.
    e. Results may need to be limited to the primary end point.
    f. Conclusion is a single brief sentence directly tied to the objective(s).
    g. Must meet word or character count limit (tricks and tips depend on organization)
    h. Revise, revise, revise with input from all authors – Eliminate unnecessary words and content.
    i. Some organizations may permit students or residents to submit abstracts without data or with partial data.
3. Abstract submission
    a. Greatly simplified by electronic submission
    b. Must meet deadline – Most websites shut down after deadline.
    c. Must carefully follow online instructions
    d. Pay careful attention to the abstract categories (e.g., clinical practice, original research) to be sure the abstract is submitted under the proper category.
    e. Word limit usually controlled by software – Difficult to cheat
    f. If platform presentations are an option, usually need to indicate consideration for platform, if that is the goal
4. Platform versus poster
    a. Platform slots are intended for presentations that have high-quality content and that are relevant and effective.
    b. Usually based on reviewer scores
    c. Many organizations may accept a platform submission as a poster presentation if it was not scored high enough to be accepted as a platform; others may just reject it.
    d. Authors must be realistic concerning the quality of their abstract when considering submission for a platform, given the meeting, audience, and likely competing research.
5. Review process
    a. Typically reviewed by three to five reviewers
    b. Review uses relatively limited scoring criteria, given the brevity of an abstract, combined with a recommendation of accept or reject.
    c. Reviews are compiled into an overall score, and recommendation is provided to the authors.
    d. There is no opportunity or time for revision and resubmission. Decisions are final.
    e. Reviewer comments may or may not be shared with the authors.
    f. Platform versus poster decisions may also be an outcome of the review process.
    g. Review process may also be used for determining abstracts to be considered for awards.
6. Poster presentation
    a. Format (size and/or style) is often specified by the organization.
    b. Many tips and tricks are available for developing effective posters. Some key issues are:
        i. Avoid wordy posters – Nobody wants to read them.
        ii. Use tables, figures, and concise bullet lists as much as possible.
        iii. Use a font size that can easily be read from about 5–6 feet (e.g., 24 point or greater).
        iv. Ease of readability is more important than aesthetics – Consider dark letters on a white background.

v. Use a logical flow from left to right and from the introduction to the conclusions.
vi. Unless required, do not reprint an abstract on a poster – It is unnecessary and uses valuable space.
vii. Most institutions have requirements to use logos – Comply with the requirements or potentially run into last-minute challenges with printing.
viii. Large-print formats have eased the production and transport of posters; however, review proofs carefully for content changes before printing.
ix. Commercial printers who will ship to the meeting site are a great alternative if last-minute challenges develop.

c. If there are walk-rounds, be fully prepared to present the key points of your poster in about 5 minutes to allow time for questions. It is important to confirm the time allotment set by each organization because this may vary.
d. Consider having small, legible versions of the poster at the poster session for those who want a copy to review. In addition, have business cards available.
e. Plan to have at least one author stay for the duration of the poster session.
f. Virtual poster sessions are very similar in submission, review, and acceptance process. Presentations are virtual and may involve the abstract only or a more detailed "poster," with interactive sessions scheduled with either random viewers or scheduled peer reviewers.

7. Platform presentation
a. Considered an honor of recognition for high-quality, innovative, or impactful work
b. Presentation is usually limited to 10 minutes, with 5–10 minutes left for questions.
c. Typical format is a brief slide presentation focused on the most important aspects of the work. Time does not allow a detailed description of all aspects of the project.
d. May involve peer review/judging if awards are involved
e. Feedback in verbal or written format is often provided to the presenter.
f. May also require a poster presentation during one of the poster sessions (varies by organization)
g. Repeated practice with coauthors, peers, and colleagues, followed by critique and revision, is highly recommended.

E. Participation in the Peer-Review Process (Eos 2011;92:233-40; PLoS Comput Biol 2006;2:e110)
1. Reasons to participate
a. Professional obligation
i. Authors "take" from the process, so they should "give back."
ii. Contribute expertise to improving the biomedical literature
b. Professional service to an organization or journal
c. Recognition, tenure, and promotion – Professional service
d. Some enjoy reviewing the "raw" product of the biomedical literature.
e. Educational opportunity for trainees
2. Reasons to decline invitation to participate
a. Conflict of interest
i. Former trainee is author
ii. Collaborator or coworker is author
iii. Financial conflict of interest with the subject
b. Lack of expertise in the subject matter
c. Lack of time to meet the deadline because of other commitments
3. Getting on the list of potential reviewers
a. Publishing in the journal
b. Being recommended by a peer to the editor
c. Being a recognized expert (nationally, internationally) in a relevant field

d. Publishing in the field in peer journals
e. Volunteering through the journal's website or a general call for reviewers – Not an option for all journals

4. Review process
   a. Invitation
      i. Normally sent by e-mail with a response link
      ii. Includes an abstract of the paper and often identifies the authors and institution
      iii. Deadline for submission of review is provided so that reviewer can gauge availability of time.
      iv. Usually a short timeline to respond to invitation
      v. If decline, most journals ask for recommendation of a peer to review (building their reviewer database)
      vi. Once accepted, online access to content is provided (manuscript, review form, instructions to reviewers, etc.).
   b. Tips for the review
      i. Biomedical literature review skills are beyond the scope of this chapter.
      ii. Review is expected to focus on scientific quality and importance of the paper.
      iii. Grammar, sentence structure, and word choices are normally better handled by the copyeditors because of their greater expertise, unless it is critical to the scientific meaning of the paper.
      iv. Connect the dots to find common errors (original research and systematic reviews).
         (a) Objective/purpose and conclusions must match.
         (b) Methods must be directly related to the objective/purpose.
         (c) Methods must be valid, widely accepted, and/or state of the art, including statistical handling of the data.
         (d) Widely accepted guidelines according to study design should be complied with (e.g., CONSORT, STROBE, PRISMA, STARD).
         (e) Every method described must have results reported.
         (f) Every result reported has to be tied to a method description.
         (g) Conclusions must be limited to and supported by the study results.
         (h) A fair and complete discussion of study weaknesses must be presented.
   c. Recommendation to editor
      i. Reason for rejection
         (a) Fatal flaw – No amount of rewriting or reanalysis of the data will make it worthy of publication – Poor-quality research or serious errors in methods
         (b) Extent of revision required is so extensive that it is equivalent to starting over
         (c) Valid research that is not important, either because of lack of relevance to the target journal or because manuscript is reporting a well-known finding with no new information
         (d) Republication of all or an extensive portion of the content (may be justifiable [e.g., portions of the methods section when it is a legitimate secondary publication of a previously published study])
         (e) Serious ethical violations
      ii. Revision (minor or major revision)
         (a) Usually for publications believed to have adequate quality and importance, but there are weaknesses that need to be corrected or addressed
         (b) Major revisions usually require a second review – The journal may ask whether you will serve as a continuing reviewer, or it may be assumed.
         (c) Minor revisions may not require rereview; depends on reasons for revision
         (d) Most journals will share the recommendation to the authors, together with the other reviewers' comments to author – Can be very instructive for future reviews

iii. Accept (without revision)
(a) Unusual with first submission and review
(b) Is a truly exceptional manuscript

d. Submission of the review
i. Electronic submission according to the journal
ii. Format is often dictated by the journal.
iii. Meet the deadline – Delayed reviews prolong the timeline and create workflow challenges for the journal.
iv. Comments to the authors
(a) Should be clear, concise, and factual. Should not be abrasive or a personal attack.
(b) Comments should be clearly referenced to the location in the manuscript (e.g., line number, page/paragraph/sentence).
(c) Are usually anonymous, but some journals provide option to be identified
(d) For manuscripts with a fatal flaw, comments can be limited to that flaw.
v. Comments to the editor
(a) Should not require extensive comments beyond those to the authors
(b) Is an opportunity to further explain the rationale for the recommendation or ethical concerns
(c) Are attributed to the reviewer, but not shared with the authors
vi. Miscellaneous
(a) Journal may ask whether you believe an editorial is needed, and a proposed author
(b) There may be a question about concerns with ethics, animal treatment, or human subjects' protection.

5. Rereview process
a. Response content
i. Cover letter detailing response to comments/requests of editor and reviewers
ii. Revised manuscript (with or without "track changes")
b. Review process
i. Should restrict comments to responses in first review. Finding an entirely new set of criticisms to original content is considered "not playing fair."
ii. Confirm that the revisions have not materially altered the meaning of other parts of the manuscript.
iii. If the authors have argued not to make a recommended revision, evaluate the validity of the response.
iv. Recommendations are the same options as for the primary review.
v. If major revisions are still needed, the editor may decide to reject, or it may undergo another round of reply and review.
vi. Deadline for the review is often shorter than for the primary review.

Table 1. Critical Care Pharmacy Services in Level 1 Critical Care Centers (Crit Care Med 2020;48:1375-82)

Patient Care
Essential: • Rounding with interdisciplinary team • Assists with informed pharmacotherapy decisions • Provision of drug information to the critical care team • Provision of drug therapy education to critical care team • Assists with prevention of inappropriate drug therapy • Provides clinical consultation for pharmacotherapeutic issues related to critical illness • Medication consults are available 24 hr/d, 7 d/wk • Provision of pharmacokinetic monitoring and therapeutic adjustments for targeted drugs • Review of medication history for determination if maintenance therapy should be continued • Medication reconciliation services for ICU patients upon admission, transfer, and discharge • Prospective review of all orders for verification of appropriateness • Patient and/or caregiver medication therapy education • Performance of independent patient assessments, e.g., pain/agitation, delirium, nutrition • Response to cardiac life support by ACLS certified pharmacists 24 hr/d, 7 d/wk • Response to other resuscitation and time-dependent emergencies in the hospital • Provision of stewardship of antimicrobials and other selected medications • Collaboration with other pharmacists within the hospital as needed • Provision of nutrition therapy plan and modifications in concert with clinical dieticians • Use of medical record for communication with other healthcare professionals • Documentation of patient care and economic impact of services • Documentation of all clinical activities • Pharmacy liaison with other interdisciplinary team members relative to medication policies, procedures, guidelines, and pathways • Utilizes pharmacoeconomic analyses to evaluate new critical care pharmacotherapeutic agents • Proactive in designing, prioritizing, and promoting new clinical pharmacy programs • Evaluation of clinical pharmacy services for stakeholder satisfaction, significance, and economic value • Participation in planning and implementation of processes for disaster and mass casualties • Majority of time dedicated to critical care services • Patient care time devoted to care of the critically ill patients • Decentralized clinical services 24 hr/d • Development of critical care pharmacist “teams” • Appropriate pharmacist-to-patient ratio based on patient acuity and complexity **Desirable:** • Provision of medication management problems documentation and progress notes daily • Preparation and presentation of drug monographs to the P&T Committee • Provision of telemedicine services if onsite critical care pharmacists are absent

Table 1. Critical Care Pharmacy Services in Level 1 Critical Care Centers (Crit Care Med 2020;48:1375-82) *(cont'd)*

Quality Improvement
Essential: • Medication safety leaders for critically ill patients • Assists with management and reduction of ADEs • Reporting of ADEs to institutional committees and national programs • Evaluation of availability of critical medications through automated dispensing cabinets • Team member in the design process for building new and remodeled critical care areas • Implements and maintains policies and procedures related to safe and effective medication use • Coordination of development and implementation of ICU-focused drug therapy guidelines, order sets, and/or care ICU pathways • Evaluation of impact of ICU-focused drug therapy guidelines, order sets, and/or care ICU pathways • Leads or provides critical care pharmacotherapy to hospital committees • Service on hospital committees • Contribution to hospital newsletter and drug monographs • Identifies and evaluates drug minimization opportunities • Identifies local quality metrics for CQI • Participates in quality assurance programs • Shares responsibility for hospital performance for quality and process measure compliance • Collaborates with other healthcare professions in preparing for accreditation • Assessment of adequacy of pharmacy space and facilities • Implementation of safety technology • Presence of contemporary medication use systems within institution • Presence of contemporary information management systems within institution **Desirable:** • Use of real-time dashboard or analytics monitoring of quality metrics and drug utilization
Research/Scholarship
Essential: • Participates as key investigator for critical care research • Representation on the IRB as applicable **Desirable:** • Active involvement with critical care pharmacotherapy research • Contributes to the pharmacy and medical literature • Report of research results at professional and scientific meetings • Participates in research design and data analysis • Secures research funding • Collaboration in multicenter research projects • Peer reviewer of the medical literature

Table 1. Critical Care Pharmacy Services in Level 1 Critical Care Centers (Crit Care Med 2020;48:1375-82) *(cont'd)*

Training/Education
Essential: • Provision of training and mentoring of pharmacy students, residents, and fellows • Supports postgraduate residencies and/or fellowship training in critical care pharmacy practice • Evaluation and documentation of trainee outcomes • Provision of education to health professional students and trainees • Development and implementation of training programs for ICU personnel **Desirable:** • Project advisor for pharmacy students, residents, and fellows • Provision of formal education at accredited educational sessions • Active participation in interdisciplinary simulation activities • Certified instructor for certification classes • Identifies and educates medical and community groups relative to pharmacist role on healthcare team
Professional Development
Essential: • Maintains master of critical care pharmacotherapy knowledge • Maintains certification in life-support courses • Seeks board certification when eligible • Involvement in nonpatient care activities • Provision of formal accredited educational sessions • Membership in critical care and pharmacy organizations • Availability of protected time for education, administrative, and research/scholarship activities • Mechanisms in place for career development

ACLS = advanced cardiac life support; ADE = adverse drug events; CQI = continuous quality improvement; ICU = intensive care unit; IRB = institutional review board; P&T = pharmacy and therapeutics.

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ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: A**
The journal Drug Intelligence and Clinical Pharmacy (now Annals of Pharmacotherapy) was the first to publish a critical care therapeutics column in 1982, which was a landmark event relative to the evolution of critical care pharmacy (Answer A). Although the other journals listed—Pharmacotherapy (Answer B), Chest (Answer C), and Heart and Lung (Answer D)—publish critical care therapeutics articles, Annals of Pharmacotherapy was the first to incorporate a critical care therapeutics column into its publication.

2. **Answer: C**
In 2001, ACCM, which exists within the organizational framework of SCCM, formed the two task forces focused on models of critical care delivery, the definition of an intensivist, and the practice of critical care medicine within three different proposed models (Answer C). Neither the Institute of Medicine (Answer A) nor ACCP (Answer B) was involved in formulating the levels of critical care delivery. Although the Clinical Pharmacy and Pharmacology Section of SCCM (Answer D) may have contributed to this document, it is not mentioned in the publication.

3. **Answer: D**
In a 2009 study by MacClaren and Bond, mortality increased in thromboembolic and infarction-related events in ICU patients without clinical pharmacy services compared with ICU patients with clinical pharmacy services: 37%, $p<0.0001$ (Answer D). Although the impact of clinical pharmacists in affecting QTc-interval prolongation (Answer A), preventable adverse drug interactions (Answer B), and drug-drug interactions (Answer C) has been evaluated, differences in mortality have not been documented. Therefore, these answers are incorrect.

4. **Answer: A**
The core knowledge areas for pharmacists caring for critically ill patients include pulmonary, cardiology, psychiatry, oncology, neuroscience, nephrology, hepatology, nutrition, gastroenterology, surgery, trauma, burn, pharmacology, transplantation, supportive care, medical emergencies, immunology, endocrinology, hematology, nephrology, toxicology, and surgery. Therefore, nephrology (Answer A) is correct. Dermatology (Answer B), rheumatology (Answer C), and obstetrics (Answer D) are not considered core knowledge areas and are therefore incorrect.

5. **Answer: C**
The landmark study documenting a decrease in preventable adverse drug reactions after the inclusion of pharmacists on interdisciplinary medical rounds was published in the Journal of the American Medical Association by Dr. Lucian Leape and colleagues (Answer C). This highly publicized article published in a mainstream medical journal by a physician remains one of the foundational studies documenting the association of critical care pharmacy services with favorable health care outcomes. The other mainstream medical journals listed, New England Journal of Medicine (Answer A), Lancet (Answer B), and Annals of Internal Medicine (Answer D), have not published similar articles on preventable adverse drug reactions after the inclusion of pharmacists on interdisciplinary medical rounds.

6. **Answer: B**
As stated, there were eight ASHP-accredited critical care pharmacy residencies in 2001. In 2021, ASHP notes 168 ASHP-accredited critical care pharmacy residencies. Assuming linear growth, the 160-residency increase over 20 years equals an increase of eight residencies per year (Answer B is correct). Although this represents significant growth, more than 2200 pharmacists would be needed to provide critical care pharmacy services, assuming 30 patients/pharmacists in the more than 67,000 adult ICU beds in the United States as of 2009. Answer A (5 residencies/year), Answer C (12 residencies/year), and Answer D (15 residencies/year) are incorrect.

7. **Answer: C**
The preferred and recommended pathway to training in critical care pharmacy is a PGY1 pharmacy practice residency, followed by a PGY2 critical care residency. This is especially true for the provision of desirable-to-optimal pharmacy services in ICUs providing level I and II services. Critical care fellowship training is an option that would also be considered preferred; however, the intent is for a greater research and academic focus. The demands of the workplace often exceed the supply of PGY2-trained critical care pharmacists. And

although many of the alternative pathways available to gaining experience, knowledge, and skills in critical care pharmacy have been successful, these are not considered preferred pathways, given the high degree of variability and inconsistency of resources to support them, therefore options A, B and D are non-preferred alternative pathways and are incorrect.

8. **Answer: C**
Even though the principles of team-based health care are all interdependent, effective communication is most tightly linked to mutual trust. Open and frank communication and the willingness to state your beliefs and challenge those of your teammates require a high level of mutual trust to keep the relationship and conversation professional and nonpersonal. Without mutual trust, communication can be more guarded, ineffective, and political. Options A, B and D are relevant principles of team-based healthcare, but not as tightly linked to effective communication, and are incorrect.

9. **Answer: B**
Although structured mentor-mentee programs can be successful, there is a greater probability of success with relationships that are voluntary and that evolve from mutual interests and a perceived opportunity to have a mutually beneficial relationship. Mentors must be willing to serve in this role, which can require a great deal of time and effort; they wish to work with mentees who are highly motivated with a track record of accepting challenges. Moreover, mentees must be responsive to feedback and teaching. Mentees seek out mentors with shared interests, a record of sharing their time and expertise, and the necessary prestige and position in the organization to promote and create opportunities for them. In an arranged relationship, it is less likely that all of these factors will come together to lead to a highly productive relationship. Mentoring relationships can exist both within and outside formal training programs like residencies and fellowships so option A is incorrect, and clinical pharmacists will often have several mentors through the different stages of their career to address different and evolving needs as they mature in their practice and scholarly activities so option D is incorrect. Finally, although mentored training programs are a viable option for the nonconventional training of critical care pharmacists, they are not an exclusive pathway so option C is incorrect.

10. **Answer: D**
Continuing pharmacy education should be included as an important strategy in a CPD PDP, and therefore options A and B are incorrect. The self-directed learner should select CPE programs that are relevant to their PDP, incorporate active learning strategies, are preferably curricular based, and are free of commercial or other bias. Continuing professional development is not an alternative to CPE; rather, it is an individualized, self-directed, continuous, and iterative process intended to address specific learning objectives developed over time by the pharmacist-learner. Continuing pharmacy education is an important component of this process, but it should not be the only learning strategy. Both CPD and CPE can include multiple learning strategies, so option C is incorrect.

11. **Answer: C**
The recently published Standards of Practice for Clinical Pharmacy, which includes a standardized process of care endorsed by all major pharmacy practitioner organizations, is intended to be applicable to any practice environment, regardless of acuity or complexity. Like other professions (e.g., medicine, nursing, physical therapy), clinical pharmacy must define and apply standards of care to create consistent expectations by all stakeholders. It is often argued that clinical pharmacists in different complex or unique practice environments cannot possibly conform to a standard of care; however, a thoughtful review of the Standards of Practice for Clinical Pharmacy reveals that it can be easily incorporated into any practice environment, and therefore options A, B and D are incorrect.

12. **Answer: A**
Clinical pharmacists practicing in the ICU have long had a broad educational role that includes students and residents in their own profession, residents and students in other professions, and colleagues on the critical care team, as well as coworkers in the pharmacy department, among the target audiences. Clinical pharmacists have used many different strategies and techniques to teach these diverse audiences across different learning environments. Although there are exceptions, the frequent or regular inclusion of patients and families in their educational activities is a more recent development. Many factors have led to this change, including a greater focus on patient- and family-centered care, greater inclusion of

patients and families as members of the team, increased demand for inclusion by patients and families, inclusion of patient satisfaction scores in pay-for-performance metrics (including pain control and understanding of medications), and greater patient awareness and interaction in the ICU with changes in sedation goals. It is anticipated that clinical pharmacists in the ICU will increasingly be directly involved in patient and family discussions and education. Options B, C and D all represent traditional audiences for clinical pharmacist educational efforts and are therefore incorrect.

13. **Answer: B**

In the context of CPD, episodic opportunities for reflection refer to spontaneous, unscheduled events that contribute to the self-assessment of learning and training needs that can be incorporated into the overall PDP. Scheduled reflection usually involves a predictable cycle like performance evaluations, annual self-evaluations, peer feedback as part of annual assessment, or a decision to schedule reflection around some set anniversary (e.g., hire dates, birthdays, end of academic year), and therefore options A, C and D are incorrect. Examples of opportunity for episodic reflection may include the challenges of managing a very difficult case, post-event debriefings for code responses, experiences with a difficult student or resident, or a request to develop a treatment guideline that is outside the clinician's usual area of expertise.

14. **Answer: B**

A methodological flaw that results in the collection and reporting of incorrect data would be considered a "fatal flaw" that no amount of rewriting or reanalysis could correct. Poor writing can be corrected with revision by the authors or during copyediting, but if the study is well conducted and has value, this would not necessarily be a reason to recommend rejection so option A is incorrect. Disagreements concerning statistical analysis are not uncommon; however, the authors may have a valid explanation or be able to revise the statistical analysis if it is not a major departure from the original intent of the study meaning option C is incorrect. It is also not uncommon for the abstract to disagree in some way with the body of the manuscript, and there is an opportunity to correct that during revisions so option D is also incorrect.